

TOULOUSE FRANCAZAL AP, France

WMO: 076310

Lat: 43.5403N

Lon: 1.3727E

Elev: 163

StdP: 99.38

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.5	-2.0	-6.9	2.1	-1.2	-5.0	2.5	-0.1	11.2	8.4	9.7	8.6	2.2	280	0.470

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	33.7	21.3	31.7	20.6	30.0	20.1	22.6	30.4	21.8	29.1	21.0	27.7	3.3	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	15.1	25.4	19.4	14.4	24.9	18.7	13.8	24.2	67.1	30.3	64.2	29.2	61.3	27.8	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	9.4	8.3	7.3	DB	-4.8	37.1	1.7	2.2	-6.0	38.6	-7.0	39.9	-7.9	41.1	-9.1	42.6
(5)				WB	-5.5	24.0	1.9	0.7	-6.9	24.5	-8.1	24.9	-9.1	25.3	-10.6	25.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.2	6.2	6.9	10.2	12.8	16.4	20.7	22.6	22.5	19.4	15.6	9.8	6.9	(6)
(7)		DBStd	6.83	3.58	3.80	3.30	2.99	3.25	3.37	3.00	3.00	3.03	3.66	3.73	3.73	(7)
(8)		HDD10.0	429	125	97	38	8	1	0	0	0	0	4	48	107	(8)
(9)		HDD18.3	1947	377	321	252	168	78	14	3	2	23	97	257	354	(9)
(10)		CDD10.0	1958	7	9	44	91	198	319	391	387	281	179	41	11	(10)
(11)		CDD18.3	436	0	0	0	1	17	84	136	131	54	13	0	0	(11)
(12)		CDH23.3	3793	0	0	1	27	153	741	1238	1167	422	43	0	0	(12)
(13)		CDH26.7	1403	0	0	0	2	31	267	499	484	116	3	0	0	(13)
(14)	Wind	WSAvg	3.3	3.2	3.6	3.8	3.8	3.4	3.3	3.0	2.9	3.0	3.1	3.0	(14)	
(15)	Precipitation	PrecAvg	654	54	47	52	65	74	62	47	50	54	53	52	(15)	
(16)		PrecMax	920	146	95	118	171	180	265	185	151	162	146	168	(16)	
(17)		PrecMin	406	2	6	0	10	20	7	2	9	0	3	3	(17)	
(18)		PrecStd	122	31	25	27	35	34	40	31	29	35	39	28	34	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.9	18.4	22.7	26.5	29.9	34.9	35.7	37.4	31.9	26.8	20.8	16.5	(19)
(20)			MCWB	12.7	11.8	13.5	16.3	18.6	21.1	21.2	21.8	20.1	18.0	15.6	12.9	(20)
(21)		2%	DB	13.9	15.7	19.7	23.4	27.0	31.6	33.5	33.9	29.8	24.4	18.6	14.8	(21)
(22)			MCWB	11.3	10.9	12.8	15.1	17.8	20.8	21.1	21.4	19.2	17.8	14.6	11.7	(22)
(23)		5%	DB	12.6	13.9	17.6	20.9	24.8	29.7	31.4	31.6	27.8	22.6	16.8	13.6	(23)
(24)			MCWB	10.5	10.0	11.7	14.0	17.2	20.1	20.5	20.8	18.4	17.1	13.9	11.1	(24)
(25)		10%	DB	11.4	12.4	15.7	18.7	22.7	27.7	29.6	29.4	25.8	21.1	15.2	12.3	(25)
(26)			MCWB	9.7	9.5	11.0	13.0	16.2	19.2	20.1	20.2	18.0	16.6	12.9	10.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.2	12.8	14.8	17.1	20.0	23.1	23.4	23.7	21.5	19.8	16.4	13.4	(27)
(28)			MCDWB	14.8	16.6	19.8	24.4	26.8	31.2	32.0	31.9	28.9	24.2	19.7	15.7	(28)
(29)		2%	WB	11.8	11.5	13.4	15.7	18.8	21.9	22.5	22.6	20.5	18.9	15.2	12.2	(29)
(30)			MCDWB	13.4	14.6	18.0	21.7	25.3	29.6	30.2	30.3	26.6	22.3	17.8	14.2	(30)
(31)		5%	WB	10.9	10.7	12.5	14.6	17.7	20.9	21.7	21.7	19.7	18.1	14.2	11.4	(31)
(32)			MCDWB	12.4	13.1	16.5	19.9	23.2	27.9	28.9	28.9	25.4	21.3	16.3	13.3	(32)
(33)		10%	WB	9.8	9.8	11.6	13.5	16.7	19.9	20.8	20.9	18.9	17.3	13.2	10.5	(33)
(34)			MCDWB	11.2	12.1	15.0	17.8	21.6	26.1	27.3	27.6	24.0	20.4	14.9	12.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.6	7.9	8.9	9.3	9.3	10.3	10.8	11.2	10.7	8.9	7.0	7.0	(35)
(36)			MCDBR	8.0	10.7	12.6	13.6	13.5	14.0	14.7	15.4	13.8	10.7	9.0	8.4	(36)
(37)			MCWBR	5.6	6.7	6.7	6.8	6.0	5.3	4.9	5.1	5.2	5.1	5.6	5.9	(37)
(38)		5% WB	MCDWB	6.9	9.3	10.1	11.9	11.4	12.3	12.1	12.6	10.9	8.1	7.7	7.5	(38)
(39)			MCWBR	5.2	6.3	5.8	6.3	5.6	5.2	4.7	4.7	4.4	4.1	5.1	5.5	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub	0.320	0.336	0.376	0.398	0.407	0.418	0.411	0.411	0.379	0.366	0.339	0.319	(41)	
(42)		taud	2.476	2.411	2.305	2.250	2.280	2.283	2.305	2.315	2.389	2.430	2.460	2.479	(42)	
(43)		Ebn at Noon	810	855	859	866	866	855	857	844	846	810	777	775	(43)	
		Edh at Noon	73	91	114	131	130	131	126	121	104	88	73	66	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.45	2.39	3.63	4.69	5.34	6.13	6.07	5.43	4.35	2.90	1.71	1.35	(44)	
(45)		RadStd	0.12	0.26	0.33	0.49	0.39	0.43	0.43	0.33	0.22	0.23	0.19	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page